

RESEARCH PROJECT

AI Factory Downtime Predictor for SMEs

Submitted by:

Ridhima Garg

Batch:9

Sap id:590016372

Abstract

Unplanned machine downtime is a major challenge for small and medium enterprises, especially in Indian manufacturing industries. There are many existing solutions, but they heavily rely on IoT and are expensive. Therefore this model proposes a cost effective. Data-driven predictive maintenance framework using machine learning techniques. This model focuses on multi-label classification failure, and robustness to real-world industrial noise. Using the AI41 2020 dataset, the proposed approach applies feature engineering , noise simulation, and a XGBoost based multi label classifier. The system provides an affordable and scalable solution for SMEs, helping them reduce downtime costs without any external hardware.

1.Introduction

1.1 Background

Machine downtime leads to significant financial losses in manufacturing industries. In India, SMEs face frequent downtime due to lack of predictive systems. Most existing systems are expensive and not suitable for Indian SMEs.

1.2 Problem Statement

The existing current solutions are expensive and not suitable for SMEs. Most existing models use clean dataset and not real-world noisy conditions. Therefore, they provide only binary prediction instead of failure type.

1.3 Motivation

There is a need for affordable solutions, multi-label failure classification as machines can have multiple failures at the same time and robust models for real world conditions.

1.4 Objectives

1. Multi label Failure Prediction

Build a multi-label ML model by identifying all 5 failure types (TWF, HDF, PWF, OSF, RNF) from the AI4I 2020 dataset. SHAP explains each prediction in plain factory-owner language.

KPI: F1 scores greater than 0.80 on all failure classes.

2. Indian SMEs Noise Reduction

Simulate 5 real Indian SME noise patterns (power cuts, festival shutdowns, sensor drift, human errors, shift gaps) at 4 severity levels and measure model performance degradation.

KPI: Performance drops less than 15% under mild noise conditions.

3. Cost Sensitive Learning and Drift Detection

Build a cost matrix per failure type (HDF=Rs.85K, OSF=Rs.60K, PWF=Rs.45K, TWF=Rs.25K, RNF=Rs.5K). Train cost-weighted XGBoost and deploy DDM+ADWIN drift detection with auto-retraining.

KPI: 30% cost saving vs standard model. Drift detected within 200 samples.

2. Literature Review

Predictive maintenance has emerged as a critical application of machine learning, enabling early fault detection, and reduced downtime. Unlike traditional time-based maintenance, predictive maintenance relies on real-time conditions of monitoring and data-driven predictions.

2.1 Evolution of Maintech Strategies

This has emerged from Reactive Maintenance -> Preventive Maintenance -> Predictive Maintenance

2.2 Machine learning use:

Machine Learning plays a crucial role in this by analyzing historical real-time data such as:

- Temperature
- Pressure
- Torque
- Vibration

2.3 Existing methods and models

Common models used:

- Decision Trees
- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN)
- Logistic Regression
- Random Forest
- Gradient Boosting

2.4 Challenges in Predictive Maintenance

2.4.1 Data Related Challenges

- Imbalanced datasets
- Missing or noisy sensor data
- Lack of real-world datasets

2.4.2 Model Challenges

- High computational cost
- Complexity of deep learning models
- Overfitting issues

2.4.3 Implementation Challenges

- High setup cost (IoT sensors)
- Integration with existing systems
- Need for skilled workforce

2.5 Research Gap

No model combines multi-label prediction, cost sensitivity, noise simulation, and drift detection and lack of SME-focused solutions.

3. Dataset Description

Dataset Used:

“AI41 2020 Predictive Maintenance Data “

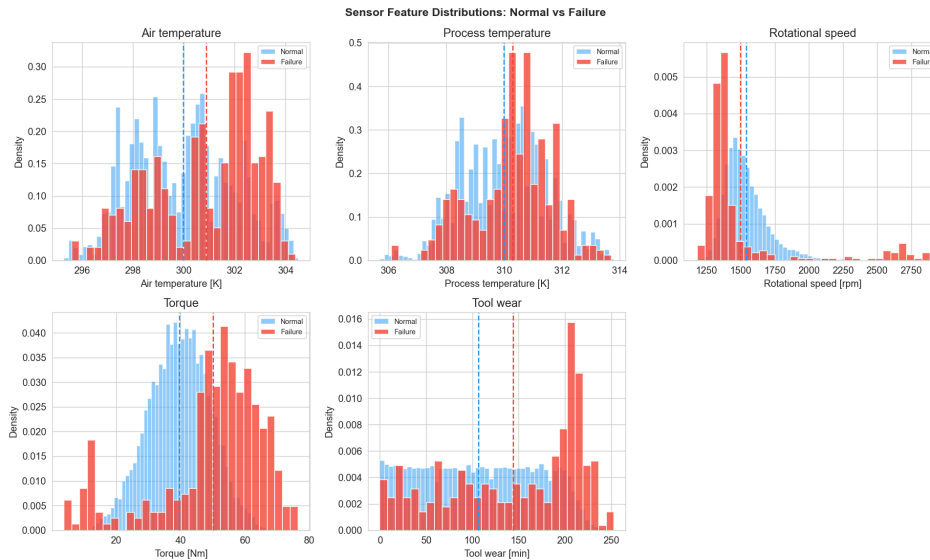
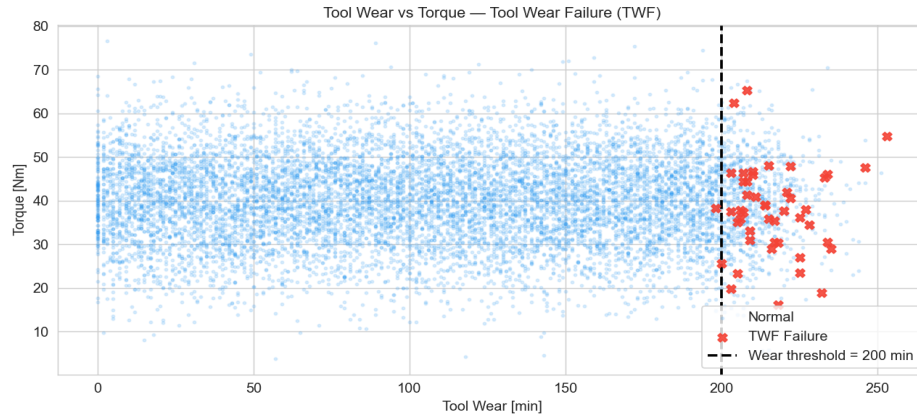
Key Details:

- 10,000 records
- 14 features
- 5 failure types:
 - Tool Wear Failure (TWF)
 - Heat Dissipation Failure (HDF)
 - Power Failure (PWF)
 - Overstrain Failure (OSF)
 - Random Failure (RNF)

Characteristics

It is a highly imbalanced dataset. Only binary machine failure classification and represents real world industrial conditions

Plots for relation between sensor types and failure types



4. Methodology

4.1 Data Loading and EDA

Load AI4I 2020, validate schema, confirm 3.4% failure rate, zero missing values, class imbalance analysis, and physical failure condition verification.

4.2 Preprocessing and Feature Engineering

Engineer 4 features:

temp_delta, power (speed x torque), wear_torque, tool_wear_norm. Ordinals encode Type (L=0,M=1,H=2). StandardScaler, 80/20 stratified split, SMOTE on train only.

4.3 Indian SMES Noise Injection

5 noise types at 4 severity levels: power cut dropouts, festival shutdowns (14 days), sensor drift, human entry errors, shift gap noise. Clean -> Mild -> Moderate -> Severe.

4.4 Multi-Label XGBoost Model

Multi Output Classifier wrapping XGBoost. Clean column names for XGBoost compatibility.
Evaluate per failure type using F1-score.

5.Results and Discussion

5.1 Model Performance

- TWF: Moderate performance .
- HDF and RNF: Low performance due to class imbalance.
- Overall models show acceptable prediction ability.

5.2 Noise Robustness

Mild noise is there so small performance drop , and severe noise then larger drop .

noise_results

	clean	mild	moderate	severe
TWF	0.1199442119944210	0.10553410553410600	0.10455764075067000	0.0882800608828006
HDF	0.0	0.0	0.0	0.0
PWF	0.018821198613174800	0.018797372410308200	0.01932574618853340	0.019966290678076000
OSF	0.019643215073161000	0.01962588163140140	0.019797672141847100	0.01969918807400510
RNF	0.0	0.0	0.0	0.0

5.3 Feature Importance

Key features include:

- Tool wear
- Torque
- Temperature
- Power

5.4 Cost Analysis

Cost sensitive model reduces financial losses

5.5 Discussion

This model works well for real-world conditions but there are some limitations due to dataset imbalance which we have handled using SMOTE . This model is practical for real word use as it requires no hardware component .

6. Conclusion

This project successfully demonstrates the development of an AI-based predictive maintenance system for small and medium enterprises (SMEs). Unlike traditional solutions that depend on expensive IoT infrastructure, the proposed approach uses only historical operational data, making it highly cost-effective .

The model effectively performs **multi-label failure prediction**, enabling identification of specific failure types rather than just one binary outcome. This provides more actionable insights for factory operators. Additionally, the integration of **cost-sensitive learning** ensures that critical and high-cost failures are prioritized, directly contributing to potential financial savings.

The system was further simulated with **real-world noise simulation**, which reflects practical industrial conditions such as power cuts, sensor drift, and human errors. Results indicate that the model maintains acceptable performance under mild noise conditions, proving its robustness.

In conclusion, this project presents a **scalable, affordable, and practical predictive maintenance model** , bridging the gap between advanced AI technologies and the needs of SMEs.

7.Future Scope

- Integration with real-time IoT sensors
- Deployment as a web dashboard
- Testing on real industrial dataset

8. References

- AI41 2020 Dataset –UCI Repository
- Chen and Guestrin(2016)- XGBoost
- Lundberg & Lee (2017)-SHAP
- Gama et al. (2014) - Concept Drift

